COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S3
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ACTIVITY NO. 14 ALGORITHM COMPLEXITY

1. PROCEDURE OUTPUT

	Best Case	Worst Case	Analysis
Algorithm 1	O(n)	O(n^2)	Bubble sort is a loop that uses Brute Force strategy. It repeatedly compares elements and swaps if it's not in the right order. Its best case is O(n) since it only needs to check if it's arranged and its worst case is O(n^2) since it repeatedly needs to check if the elements are in order. Since it repeatedly compares the input, the best case is faster than the worst case. Just like the other previous activities, it still depends on the computer on how fast its runtime is.
Algorithm 2	O(n)	O(n^2)	Insertion sort uses a while loop inside of the for loop. The best case is O(n) if the array is sorted the outer loop “for” runs n times the inside loop will not run. The worst case is O(n^2) for every element processed by the outer for loop, the inner while loop must run until it's done. As the input size increases, the best case is much faster than the worst case. It also depends on the computer how fast the runtime.

Table 14-1. Best- and Worst-Case Analysis using Theoretical Tools

	Algorithm 1 (Bubble Sort)		Algorithm 2 (Insertion Sort)	
Best Case	**Input size (n)**	**Output (us is microseconds)**	**Input size (n)**	**Output (us is microseconds)**
	100	0.4 us	100	0.4 us
	1000	1.9 us	1000	2.8 us

	10000 | 17.7 us	10000 | 25.4 us
Worst Case	**Input size (n)** / **Output (us is microseconds)** 100 | 24.2 us 1000 | 2311.6 us 10000 | 227374 us	**Input size (n)** / **Output (us is microseconds)** 100 | 14.3 us 1000 | 1232.7 us 10000 | 113913 us

Table 14-2. Best- and Worst-Case Analysis using Experimental Tools

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

TIME COMPLEXITY

Case	Linear Search	Binary Search
Best Case	O(1)	O(n log n)
Worst Case	O(n)	O(n)

SPACE COMPLEXITY

Linear Search	O(1)
Binary Search	O(1)

Input Size (N)	Elapsed Time for Binary Search (Worst-Case)	Elapsed Time for Linear Search (Worst-Case)
1,000	0.035 microseconds	1.51 microseconds
10,000	0.042 microseconds	15.34 microseconds

100,000	0.049 microseconds	164.17 microseconds

ANALYSIS:
This results shows in the table that Binary Search is faster than Linear Search as the input size increases. Binary Search has a worst-case time complexity of O(log n) because it divides the search range in half during each step, while Linear Search has a worst-case time complexity of O(n)because it may need to check every element. Both algorithms have a space complexity of O(1) because they use a constant amount of extra memory. Based on the measured times, Linear Search takes longer as the number of elements increases, while Binary Search shows only a small increase in execution time.

CONCLUSION:	
CABUG:	In this activity, we see what's faster in searching algorithms, we see that Binary Search is faster than Linear Search, especially when dealing with large amounts of data. In the sorting algorithm we also see that Bubble sort was faster than insertion sort as the input size increases. My partner and I had different output for microseconds because it depends on the computer runtime.
SALCEDO:	We were able to finish the Hands-on Activity. I was able to understand how each search and sorting algorithm affected the time measure based on their inputs. This activity reminds me of the previous module, Module 11. Overall, the activity was understandable and created new learning regarding the algorithm complexity.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Analysis**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **We used ChatGPT for the analysis and our sentences that we're using are ours only.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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